

Assignment #2-Final Phase

Project description document

Under Supervision

Dr. Mohamed Al-Ramly

By Students: S23

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Our Repo Link 

<https://github.com/Abdo2000-1/Assignment-2.git>

Our YouTube Video Link  :

<https://youtu.be/A67IX7vEtCU>

Who Did which :

Abdo:	Tasks 4, 7, 10 and Bonus Features and the Describing PDF Document and the Diagrams and The YouTube Video and Light/Dark Mode feature
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Adham :	Tasks 2, 5, 8
Moaz :	Tasks 3, 6, 9
Hamza:	Tasks 12, 13, 14 and some maintenance and GUI in phase 1

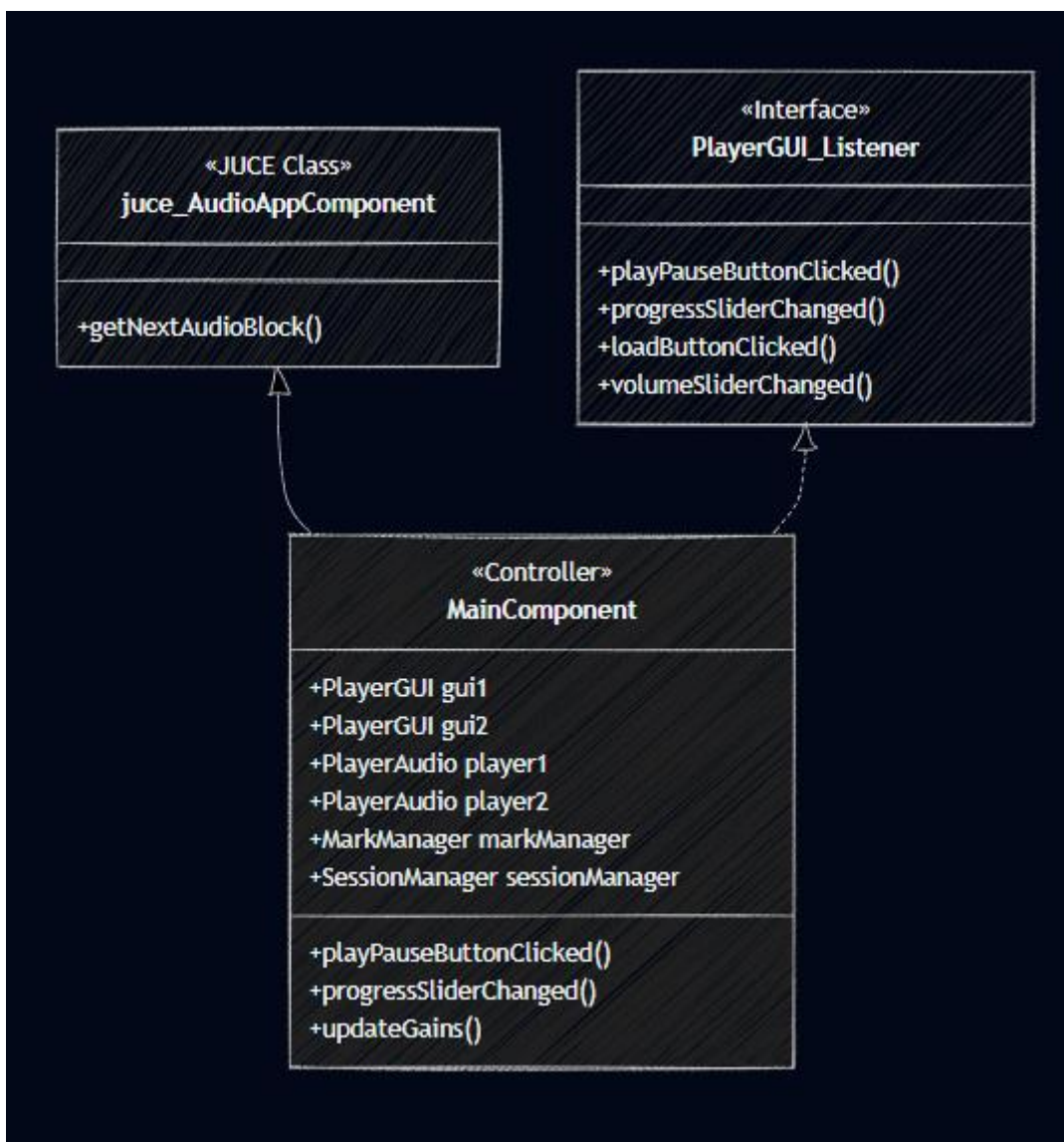
The project effectively utilizes the following core OOP concepts:

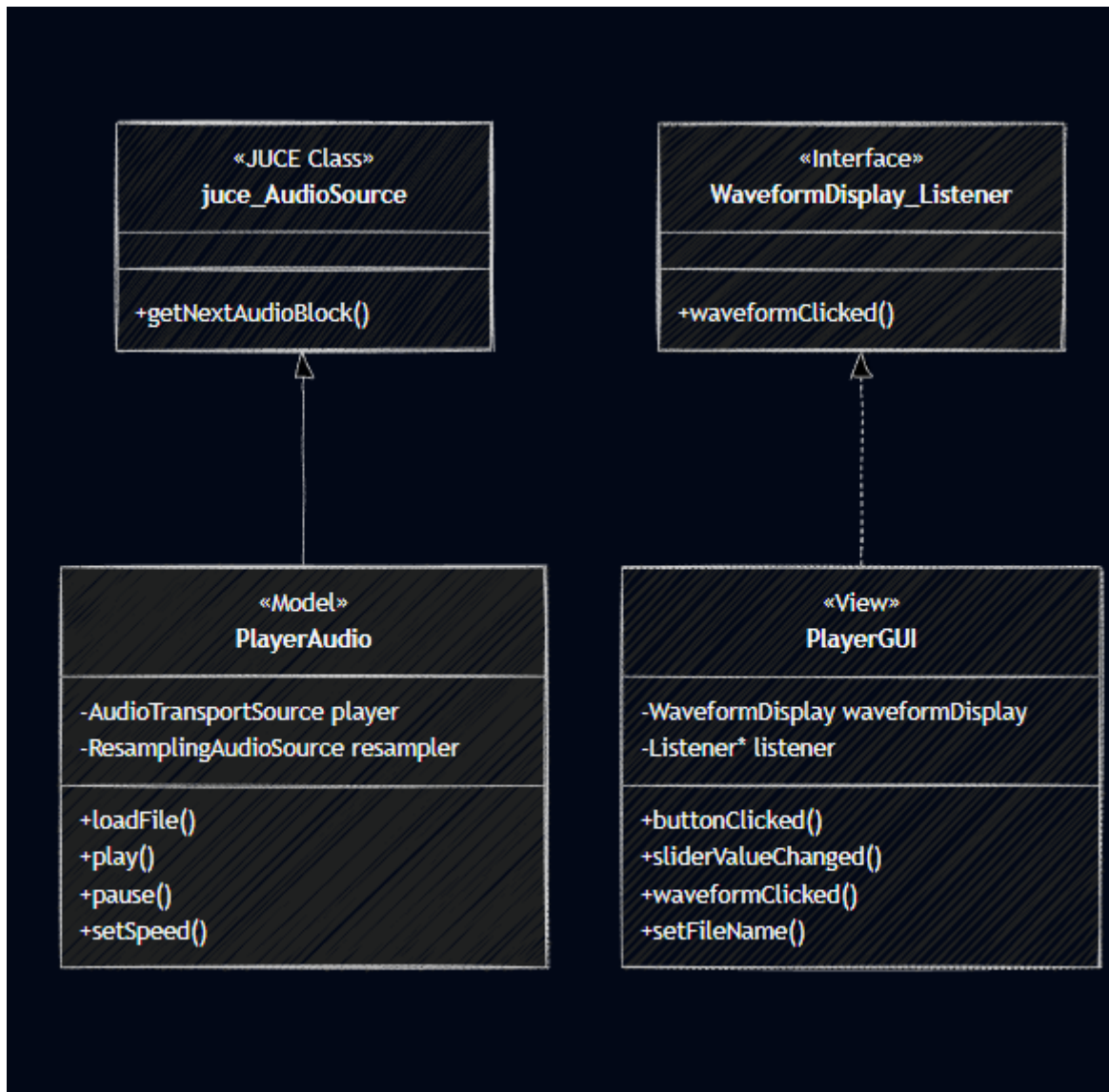
- **Encapsulation:** This is achieved by grouping related data (e.g., player, reader, metadata, volumeBeforeMute) and methods (play(), setGain(), toggleMute()) within the **PlayerAudio** and **PlayerGUI** classes. Private members hide the internal state and implementation details from external access.
- **Abstraction:** The user interface classes (MainComponent and PlayerGUI) interact with the complex audio engine (PlayerAudio) through a simple public interface (loadFile(), play(), setPosition(), setGain()). The GUI doesn't need to know the specifics of format reading, resampling (juce::ResamplingAudioSource), or transport control.
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- **Inheritance& Polymorphism:**

- **MainComponent** inherits from **juce::AudioAppComponent** to gain essential audio functionalities (like **prepareToPlay**, **getNextAudioBlock**).
- **PlayerAudio** inherits from **juce::AudioSource** to be able to feed audio data to the mixer.
- The listener pattern (e.g., **PlayerGUI::Listener**) is implemented, where **MainComponent** inherits from **PlayerGUI::Listener** to handle all button and slider events.

Class Diagrams :



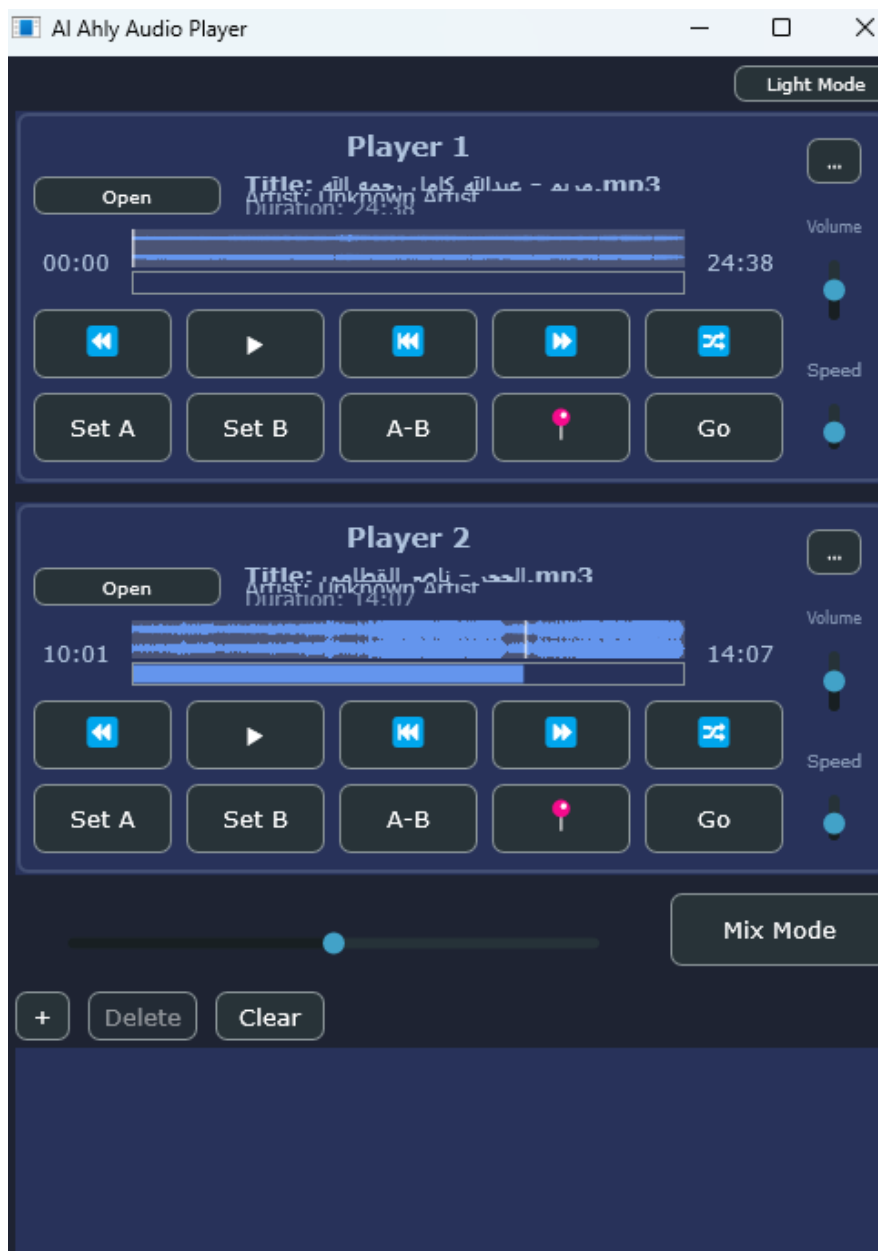






Our Ui Design







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.gitignore	commit	3 days ago
Main.cpp	7/11/2025	6 hours ago
MainComponent.cpp	7/11/2025	6 hours ago
MainComponent.h	7/11/2025	6 hours ago
MarkManager.cpp	Add marks	19 hours ago
MarkManager.h	Add marks	19 hours ago
NewProjectJucer	7/11/2025	6 hours ago
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